# Matthew Bronstein

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GitHub | LinkedIn | Portfolio

## ***Summary***

Full stack engineer and technical lead with 3+ years building and operating a computer-vision platform for warehouse customers, from React interfaces and TypeScript services to Docker-based CI/CD and on-prem/edge deployments. Promoted from frontend developer to Head of Product and Technology within three years, owning architecture, customer deployments, and cross-functional delivery. Former professional orchestral musician, bringing discipline and clear communication to high-stakes work.

## ***Technical Skills***

**Languages and Web:** TypeScript/JavaScript, React, Redux Toolkit, Next.js, Tailwind CSS, Node.js, Bun, Express, REST, GraphQL, MQTT, Python (working knowledge)

**Data and Infrastructure:** PostgreSQL, Prisma, Docker, GitHub Actions (CI/CD), Cloudflare (Zero Trust, Tunnels, R2), Tailscale, Linux (Ubuntu, Debian, Arch), RTSP (go2rtc), WorkOS, Twilio, OpenAI/Anthropic APIs, MCP

## ***Experience***

**Arvist AI** | Chicago, IL (Remote)

**Head of Product and Technology** | 10/2026 - Present

- Lead a team of 8 developers while owning platform architecture and 10+ customer deployments, coordinating regularly with the business team
- Serve as technical point of contact for prospective customers, supporting technical diligence and IT/security requirements reviews

**Lead Platform Engineer** | 01/2025 - 10/2026

- Own end-to-end architecture of the microservice platform (containerized services communicating over MQTT and REST) and the database (schema design, Prisma migrations), running 10+ customer on-prem/edge deployments as first responder for on-site issues
- Provision and harden customer edge servers (Ubuntu/Debian/Arch, BIOS/UEFI with Secure Boot and TPM, LUKS with Clevis auto-unlock, secured SSH) and documented the process for consistent hand-off
- Planned and shipped "Quality," a new flagship product now live at 9 customer locations, coordinating web and ML teams from product handoff through implementation
- Built a two-way WMS data sync so customer systems feed Quality inspections and receive results, unblocking it by routing customer requests to on-prem services through Cloudflare tunnels with service tokens

- Designed remote access using cloudflared and Cloudflare SSH applications, giving instant access to every customer site without needing IP addresses; sole owner of all customer cloud configuration
- Designed a hybrid media storage system, configurable per deployment by environment variable, that saves media to local device storage or the cloud to match each customer's security requirements
- Built GitHub Actions CI/CD workflows that build, tag, and publish Docker images to GHCR with changelogs from conventional commits, enabling independent per-service versioning; lead code review and release readiness
- Implemented session-based authentication; created and maintain public API and MQTT documentation (Docusaurus, OpenAPI)

**Full Stack Developer** | 01/2024 - 01/2025

- Built real-time interfaces surfacing computer-vision detections, alerts, and metrics to warehouse users, integrating with services over REST and MQTT
- Developed a safety module feature that validates detected incidents by time, day, camera, and zone and notifies configured users
- Built a yard management feature tracking truck arrivals and departures, with Slack alerts, duplicate-detection validation to prevent false positives, and reviewable RTSP image/video capture
- Migrated the entire backend from JavaScript to TypeScript, reducing type-related runtime errors to near zero, and introduced Prisma for schema management and versioned migrations

**Frontend / React Developer** | 06/2023 - 01/2024

- Built a canvas-based zone-drawing tool for ML detection zones and an object-tracking carousel drawing vector paths between an object's previous positions
- Improved performance through component memoization and refactoring; contributed Python backend changes

## ***Education***

**Northwestern University** | Full Stack Web Development Certificate | 11/2022

**Roosevelt University, Chicago College of Performing Arts** | B.M., French Horn Performance